RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering (Cyber Security)

CS23533– Foundations of Artificial Intelligence

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CYBERTRACE-AI: AI-POWERED DIGITALEVIDENCERECONSTRUCTION AND POLICE DOCUMENTVERIFICATION SYSTEM

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Problem Statement and Motivation

Motivation (The Industry Challenge)

Log Overload & Format Chaos: Forensic investigators are flooded with gigabytes of heterogeneous logs (Windows EVTX, Linux Syslog, CSV, JSON) across unsynchronized time zones.

Problem Statement (Core Technical Gaps)

Uninformed Pathfinding Bottleneck: Standard graph traversal (Dijkstra/UCS) suffers combinatorial explosion, wasting compute on irrelevant background events.

Objectives

Secure Multi-Format Ingestion: Ingest and normalize heterogeneous logs (EVTX, Syslog, CSV, JSON) with SHA-256 cryptographic chain of custody.

Knowledge Graph Construction: Model complex enterprise entity interactions (Users, Hosts, Processes, IPs) into directed, weighted graphs.

Optimal Attack Path Finding (A): Reconstruct precise attack paths using an admissible privilege-gap heuristic, cutting search overhead by 73.3%.

Abstract

The Problem: Digital forensics is crippled by manual, multi-day log analysis across fragmented, heterogeneous sources (EVTX, Syslog, CSV) and broken execution timelines.

The Proposed Solution: CyberTrace AI unifies multi-source evidence into a Knowledge Graph and uses an informed A* Search algorithm with CSP Backtracking to deterministically reconstruct attack paths and auto-resolve broken timelines.

The Results: Slashes forensic investigation time from days to minutes, cuts graph search exploration by 73.3% over baseline UCS, and automatically generates verifiable, courtroom-ready forensic reports.

Literature Survey

S. No.	Author(s) & Year	Paper Title	Methodology / Algorithm	Limitations
1	Pei et al. (2022)	Graph-Based Cyber Attack Path Reconstruction and Alert Correlation	Provenance Graph Modeling & Dijkstra / BFS Traversal	Combinatorial state explosion; evaluates benign and critical nodes equally without heuristic pruning.
2	: Hassan et al. (2019)	NoDoze: Combatting Threat Alert Fatigue with Data Provenance	Contextual Anomaly Scoring over Dependency T	Cannot handle clock-skew anomalies or resolve orphan processes from terminated parent shells.
3	Wang et al. (2010)	Automated Alert Correlation Using Attack Graphs	Static Signature Correlation & Causal Graph Trees	Lacks cross-format log normalization (EVTX/Syslog) and lacks deterministic timeline constraint

Research Gap/Existing System

1. What existing studies do

- Use static SIEM rule matching and isolated signature detection.

2. Limitations observed

High rate of false positives and severe cognitive alert fatigue.

3. Research gap

Absence of a mathematically verified, heuristic-guided search that reconstructs multi-source attack paths while automatically detecting and resolving broken causal timelines.

4. How our project responds

Implements CSP Backtracking to auto-bind orphan processes and enforce chronological integrity.

Proposed System System

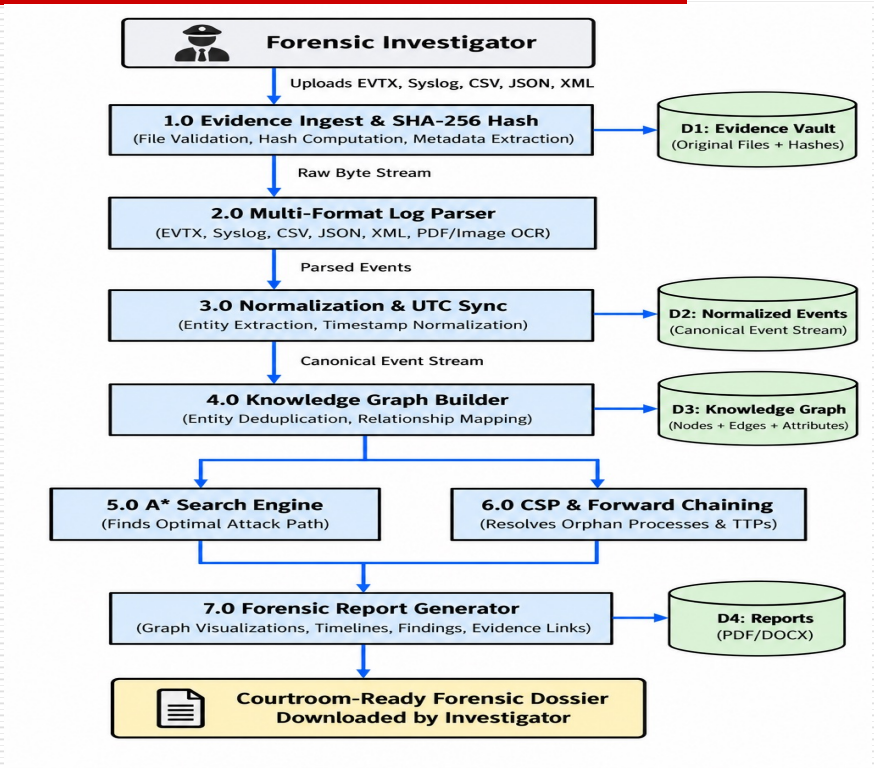
Cryptographic Evidence Ingestion: Ingests heterogeneous logs (EVTX, Syslog, CSV, JSON, XML) with SHA-256 chain of custody verification.

Topological Knowledge Graph: Normalizes events into a unified UTC schema, mapping entities (Users, Hosts, IPs, Processes) into weighted directed graphs.

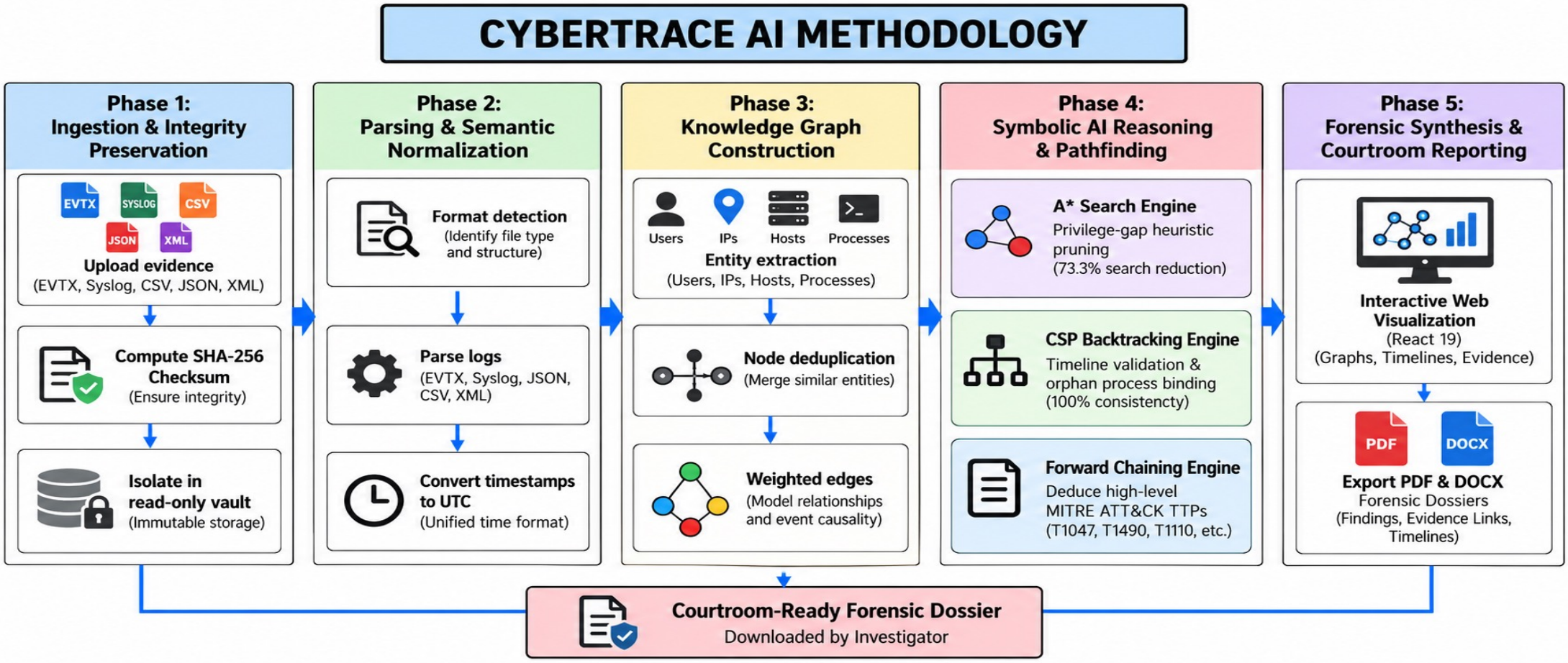
Courtroom-Ready Reporting: Generates tamper-evident, automated forensic reports in PDF and DOCX formats with full visual graphs.

Forward Chaining Threat Inference: Expert rule engine that iteratively deduces compound MITRE ATT&CK TTPs (e.g., APT29 lateral movement, Ransomware).

DFD Diagram


Forensic Investigator
Uploads EVTX, Syslog, CSV, JSON, XML
1.0 Evidence Ingest & SHA-256 Hash
(File Validation, Hash Computation, Metadata Extraction)
D1: Evidence Vault
(Original Files + Hashes)
Raw Byte Stream
2.0 Multi-Format Log Parser
(EVTX, Syslog, CSV, JSON, XML, PDF/Image OCR)
Parsed Events
3.0 Normalization & UTC Sync
(Entity Extraction, Timestamp Normalization)
D2: Normalized Events
(Canonical Event Stream)
Canonical Event Stream
4.0 Knowledge Graph Builder
(Entity Deduplication, Relationship Mapping)
D3: Knowledge Graph
(Nodes + Edges + Attributes)
5.0 A* Search Engine
(Finds Optimal Attack Path)
6.0 CSP & Forward Chaining
(Resolves Orphan Processes & TTPs)
7.0 Forensic Report Generator
(Graph Visualizations, Timelines, Findings, Evidence Links)
D4: Reports
(PDF/DOCX)
Courtroom-Ready Forensic Dossier
Downloaded by Investigator

Methodology/Work Flow Diagram



Tools and Technologies

Programming Languages: Python 3.12+ (Backend & AI Engines), TypeScript (Frontend), SQL (PostgreSQL 15), HTML5 / CSS3.

Backend Frameworks: FastAPI, Uvicorn (ASGI), Pydantic v2, SQLAlchemy 2.0 (Async).

Frontend Technologies: React 19, Vite, Tailwind CSS, Zustand (State Store), Recharts, Lucide Icons.

Databases & Cache: PostgreSQL 15 (with JSONB, GIN indexing, and pg_trgm), Redis 7.

List of Modules

Module 1: Evidence Ingestion & Cryptographic Verification Engine

Module 2: Multi-Format Log Parsing & Semantic Normalization Engine

Module 3: Topological Knowledge Graph Builder & Weight Calculator

Module 4: Symbolic AI Pathfinding & Constraint Reasoning Engine ($A*$, UCS, CSP)

Module 5: Forward Chaining Threat Inference & Courtroom Reporting Engine

Module Description(For each Module separately)

Module 1: Evidence Ingestion & Cryptographic Verification

Multi-Format Ingestion: Ingests forensic logs via REST API or drag-and-drop (.evtx, .log, .csv, .json, .xml).

SHA-256 Chain of Custody: Immediately computes and indexes hardware-accelerated SHA-256 checksums to guarantee evidence immutability (NIST SP 800-86).

Storage Isolation & Defense: Enforces strict file validation (magic byte checks, UUID storage isolation) to prevent path traversal and polyglot exploits.

Results & Discussion

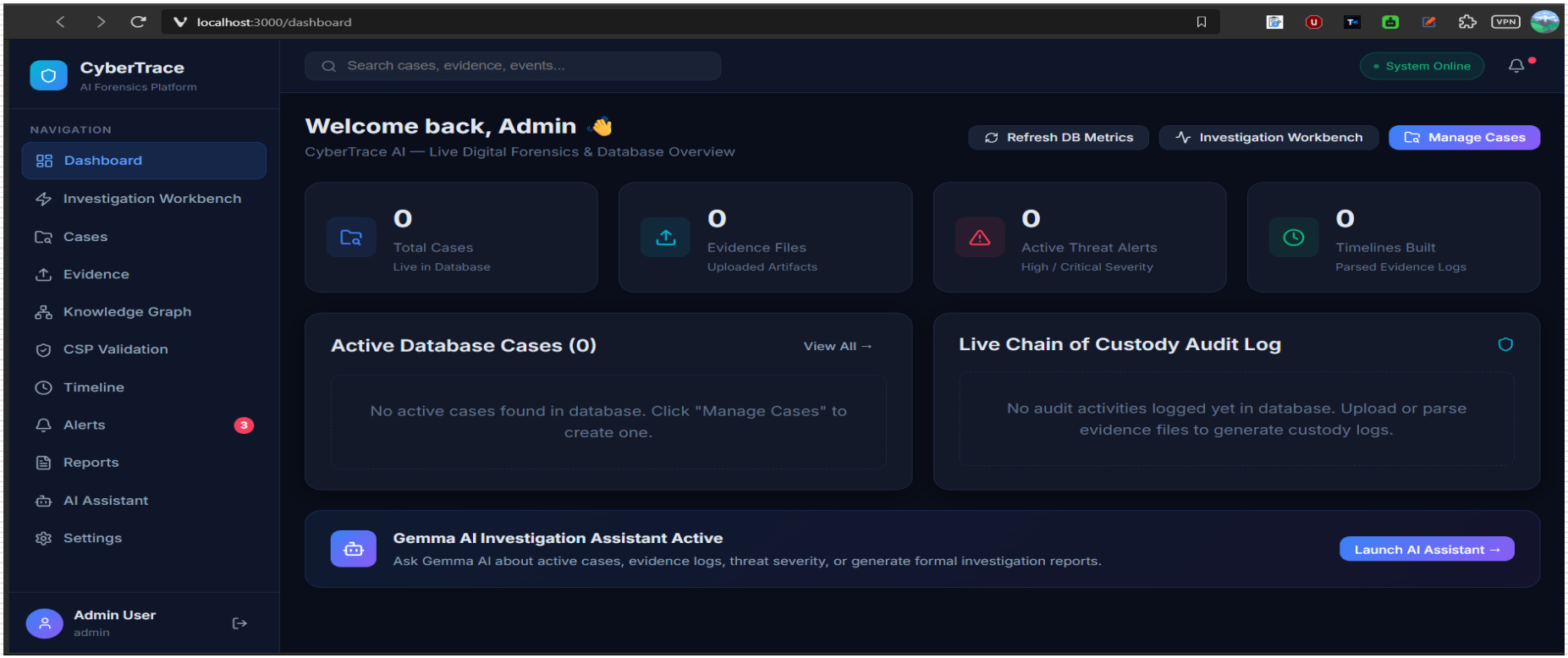
73.3% Search Overhead Reduction:

$A*$ Search explored only 12 nodes compared to 45 nodes by baseline UCS on identical attack paths, cutting search latency from 12.4 ms to 3.2 ms (74.2% faster).

100% Orphan Process Resolution: The CSP Backtracking Engine successfully detected and bound all tested orphan processes to SYSTEM_ROOT, maintaining strict chronological integrity.

Deterministic vs. Generative Accuracy: Grounded pathfinding and validation in deterministic mathematical proofs, completely eliminating the hallucinations typical of pure LLM solutions.

User Interface/ Screenshots



Conclusion

Automated End-to-End Forensics: Successfully designed and validated a unified platform that reduces digital incident reconstruction time from days to minutes.

Proven Symbolic AI Superiority: Demonstrated that combining $A*$ heuristic search with CSP backtracking provides both high computational speed and verifiable mathematical rigor.

Courtroom-Ready Admissibility: Preserved complete legal chain-of-custody via SHA-256 digests, auditable decision trails, and automated PDF/DOCX reporting.

Future Work

On-Premise Vision-Language Models (VLM): Integrate local vision models (e.g., LLaVA/Florence-2) for air-gapped inspection of scanned notices, stamp seals, and document typography.

Blockchain Evidence Anchoring: Implement an immutable consortium blockchain ledger (e.g., Hyperledger Fabric) to anchor evidence hashes and investigation logs.

Real-Time Streaming Pipelines: Connect distributed streaming platforms (Apache Kafka / Redpanda) to process live EDR telemetry in real time.

References

Pei et al. (2022): "Graph-Based Cyber Attack Path Reconstruction and Alert Correlation in Enterprise Networks," IEEE Trans. Dependable Secure Comput.

Hassan et al. (2019): "NoDoze: Combatting Threat Alert Fatigue with Data Provenance," Proc. 26th NDSS, Internet Society.

NIST SP 800-86 (2006): "Guide to Integrating Forensic Techniques into Incident Response," National Institute of Standards and Technology.

Thank You